A 2-D Unstructured Finite-Volume Compressible Navier–Stokes Solver: Implementation, MPI Parallelisation and Benchmark Results

`fvm2d` — benchmark submission

August 28, 2026

Abstract

This report documents `fvm2d`, a cell-centred finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, written from scratch in C++17 with MPI domain decomposition. The mesh is imported from CGNS (including multi-zone unstructured meshes), partitioned with METIS k-way partitioning on the cell adjacency graph, and distributed so that each rank stores only its owned cells plus one layer of ghost cells. The spatial discretisation is second order: least-squares primitive-variable gradients with a Venkatakrishnan limiter, an HLLC approximate Riemann solver with a shock-activated blend towards Rusanov, and a directionally corrected viscous flux built from primitive gradients. Steady cases are marched with a CFL-ramped local pseudo-time implicit scheme whose inner linear system is relaxed with symmetric Gauss–Seidel (LU-SGS-style simplified Jacobian); the unsteady cylinder case uses BDF2 dual time stepping with a true outer physical-time loop and frozen histories during the inner iterations. Of the 8 required cases, 8 produced a result package and 8 satisfy the benchmark convergence or statistical-periodicity criterion; the physics sanity gate in `report/sanity_checks.json` reports all checks passing. Headline results are a numerical drag of $8.7683 \cdot 10^{-4}$ for the incompressible-limit inviscid NACA0012 (exactly zero in theory), $C_d = 2.017$ for the steady Re 20 cylinder, and a Strouhal number of 0.180 with mean drag 1.229 for the Re 200 vortex street.

Contents

1 Introduction

The benchmark asks for a complete, self-contained CFD project: a compressible Navier–Stokes solver on unstructured two-dimensional meshes, distributed with MPI, driven only by the supplied JSON case files, and validated on eight cases — six NACA0012 configurations (inviscid and laminar at $M_\infty = 0.15$, 0.8 and 2.0) and two circular-cylinder configurations (steady at $Re = 20$ and an unsteady vortex street at $Re = 200$).

Nothing in the solver branches on a case name. The case JSON is the single source of truth: mesh path, physics mode, gas constants, freestream state, reference quantities, the mapping from mesh boundary families to boundary-condition types, and every run control are read from it. The two supplied meshes differ structurally — the NACA mesh is a single mixed triangle/quadrilateral zone with families `bc-2`/`bc-4`, while the cylinder mesh consists of two zones joined by 1-to-1 grid connectivity with families `WALL`/`FAR` — and both are handled by the same generic reader.

The remainder of the report follows the structure of the task specification: governing equations, mesh handling, spatial discretisation, boundary conditions, time integration, MPI parallelisation, reproducibility, results for every case, parallel validation, and an honest discussion of the limitations.

2 Governing Equations and Nondimensionalisation

2.1 Conservation form

The solver advances the conservative state

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T$$

according to

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with the inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

Both are implemented in `projectedEulerFlux` (`src/physics/Flux.cpp`), which directly evaluates the normal projection $\mathbf{F}^i n_x + \mathbf{G}^i n_y$.

2.2 Thermodynamics

The gas is calorically perfect with configurable γ, R and Pr taken from the case file (`src/core/GasModel.h`):

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}\left(u^2 + v^2\right), \qquad a = \sqrt{\gamma p/\rho}, \qquad T = \frac{p}{\rho R}, \qquad c_p = \frac{\gamma R}{\gamma - 1}.$$

Every thermodynamic relation used anywhere in the solver goes through this one class, which is the extension point for a more general equation of state.

2.3 Viscous terms

The Newtonian stress tensor and the Fourier heat flux are

$$\tau_{xx} = 2\mu u_x - \tfrac{2}{3}\mu\left(u_x + v_y\right), \quad \tau_{yy} = 2\mu v_y - \tfrac{2}{3}\mu\left(u_x + v_y\right), \quad \tau_{xy} = \mu\left(u_y + v_x\right), \quad \mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr},$$

so that

$$\mathbf{F}^v \cdot \mathbf{n} = \begin{bmatrix} 0 \\ \tau_{xx} n_x + \tau_{xy} n_y \\ \tau_{xy} n_x + \tau_{yy} n_y \\ \left(u\tau_{xx} + v\tau_{xy} + kT_x\right) n_x + \left(u\tau_{xy} + v\tau_{yy} + kT_y\right) n_y \end{bmatrix}.$$

Inviscid cases are obtained by setting $\mu = k = 0$; no separate code path is used.

2.4 Nondimensionalisation and reference quantities

The case files already supply a consistent nondimensional state ($\rho_\infty = 1$, $|\mathbf{u}_\infty| = 1$, $R = 1$, and p_∞ chosen so that $M_\infty = |\mathbf{u}_\infty|/\sqrt{\gamma p_\infty/\rho_\infty}$), so the solver works directly in those units and performs no further scaling. The consistency of M_∞ with $(\rho_\infty, p_\infty, |\mathbf{u}_\infty|)$ is verified at load time and a mismatch is a hard error. For laminar cases the constant dynamic viscosity is set from the requested Reynolds number,

$$\mu = \frac{\rho_\infty |\mathbf{u}_\infty| L_{\mathrm{ref}}}{Re},$$

with L_{ref} = `reference.reynolds_length`. Force coefficients use the freestream dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty|\mathbf{u}_\infty|^2$ and the case reference area A_{ref}:

$$C_d = \frac{F_x \cos\alpha + F_y \sin\alpha}{q_\infty A_{\mathrm{ref}}}, \qquad C_l = \frac{-F_x \sin\alpha + F_y \cos\alpha}{q_\infty A_{\mathrm{ref}}}, \qquad C_{m,z} = \frac{M_z}{q_\infty A_{\mathrm{ref}} L_{\mathrm{ref}}}.$$

3 Meshes, Boundary Tags and Case Inputs

3.1 CGNS import

`src/mesh/CgnsReader.cpp` opens the file with the CGNS mid-level library and discovers everything from the file itself: the number of zones, the element sections (`TRI_3`, `QUAD_4`, `BAR_2` and `MIXED` are supported), the boundary-condition patches, and the family name attached to each patch. Boundary patches are edge- or element-located point ranges/lists; each referenced element is looked up in the zone's bar-element table to recover the two end nodes.

Multi-zone meshes are welded into a single unstructured mesh. The declared 1-to-1 `GridConnectivity` vertex lists are used first (a union-find over the concatenated node numbering), followed by a geometric fall-back that matches coincident nodes on any interface edge that the file left undeclared. For `CylinderB1.cgns` the declared lists weld 320 node pairs and no geometric fall-back is required, reducing the $7079 + 3236 = 10315$ zone nodes to 9995 mesh nodes; the solver logs both counts at start-up in each `stdout.log`.

3.2 Geometry

Cells are stored as general polygons in CSR form and are reoriented counter-clockwise if necessary. Areas and centroids use the polygon (shoelace) formulas; face normals are $\mathbf{n} = (\Delta y, -\Delta x)/\|\cdot\|$ for the edge (n_0, n_1) of a counter-clockwise cell, which points out of the left cell, and the face "area" in 2-D is the edge length. A geometric consistency check is run at start-up on every rank: the sum of the outward face-area vectors of every owned cell must vanish. The observed maximum relative closure error is below 10^{-14} on both meshes, and a value above 10^{-10} aborts the run.

3.3 Face structure and adjacency

A hash table over sorted node pairs builds the unique face list. A face seen twice is interior; a face seen once must be covered by a boundary-edge section, otherwise the mesh is rejected with a diagnostic that prints the offending edge coordinates. This same face list is the cell adjacency (dual) graph handed to METIS.

Table 1: Required cases, meshes and global mesh sizes as reported by the solver in `metadata.json`.

case id	mesh	mode	M_∞	Re	cells	faces	run type
naca0012_m015_inviscid	NACA0012_H2.cgns	inviscid	0.15	–	20816	36498	steady
naca0012_m080_inviscid	NACA0012_H2.cgns	inviscid	0.8	–	20816	36498	steady
naca0012_m200_inviscid	NACA0012_H2.cgns	inviscid	2	–	20816	36498	steady
naca0012_m015_laminar_re5000	NACA0012_H2.cgns	laminar	0.15	5000	20816	36498	steady
naca0012_m080_laminar_re5000	NACA0012_H2.cgns	laminar	0.8	5000	20816	36498	steady
naca0012_m200_laminar_re5000	NACA0012_H2.cgns	laminar	2	5000	20816	36498	steady
cylinder_m010_laminar_re20	CylinderB1.cgns	laminar	0.1	20	10185	20180	steady
cylinder_m010_laminar_re200	CylinderB1.cgns	laminar	0.1	200	10185	20180	transient

The NACA0012 mesh has a circular farfield at $r = 80$ chords and 404 wall faces on the aerofoil; the cylinder mesh has a circular farfield at $r = 200$ diameters and 100 wall faces on the cylinder (both counts are the number of rows written to `surface.csv`).

The boundary-family mapping is taken from the case file: for the NACA cases `bc-2` is the farfield and `bc-4` is the aerofoil (slip wall for inviscid runs, no-slip adiabatic wall for laminar runs); for the cylinder cases `FAR` is the farfield and `WALL` is the cylinder. A family present in the mesh but missing from the case mapping is a hard error.

4 Spatial Discretisation

4.1 Cell-centred finite volume

Integrating the conservation law over cell c of area V_c gives

$$V_c \frac{d\mathbf{U}_c}{dt} = -\mathbf{R}_c, \qquad \mathbf{R}_c = \sum_{f \in \partial c} \left[\hat{\mathbf{F}}^i_f - \hat{\mathbf{F}}^v_f \right] s_{c,f}\, A_f,$$

where A_f is the edge length and $s_{c,f} = \pm 1$ orients the stored face normal outward from c. Because each face is visited once and its flux is added to the left cell and subtracted from the right cell, the scheme is discretely conservative by construction (`src/solver/Residual.cpp`).

4.2 Second-order reconstruction

Primitive variables $\mathbf{W} = (\rho, u, v, p)$ are reconstructed linearly,

$$\mathbf{W}_f = \mathbf{W}_c + \boldsymbol{\phi}_c \odot \left(\nabla \mathbf{W}_c \cdot (\mathbf{x}_f - \mathbf{x}_c) \right),$$

with inverse-distance-weighted least-squares gradients,

$$\nabla W_c = \mathbf{M}_c^{-1} \sum_j w_j \, \Delta\mathbf{x}_j \, (W_j - W_c), \qquad \mathbf{M}_c = \sum_j w_j \, \Delta\mathbf{x}_j \Delta\mathbf{x}_j^T, \qquad w_j = \|\Delta\mathbf{x}_j\|^{-2}.$$

The stencil contains the face neighbours (owned or ghost) *and* the boundary states evaluated at boundary-face centroids, which keeps the wall-normal gradient — and therefore the skin friction — accurate in the first cell layer. Because the geometry is fixed, the 2×2 inverse and the per-neighbour coefficient vectors are precomputed once in `LocalMesh::buildLeastSquares`. If $\mathbf{M}_c$ is numerically singular the cell falls back to a zero gradient (first order); this never occurred in any production run reported here.

Reconstruction is active in every production run: `spatial_order_claimed` is 2 in all result directories and the limiter/reconstruction code path is the only one compiled into the residual for `numerics_required.spatial_order`= 2.

4.3 Limiter and positivity

The default limiter is Venkatakrishnan's smooth limiter applied per primitive variable,

$$\phi(\Delta_1, \Delta_2) = \frac{1}{\Delta_2} \frac{\left(\Delta_1^2 + \epsilon^2\right) \Delta_2 + 2\Delta_2^2 \Delta_1}{\Delta_1^2 + 2\Delta_2^2 + \Delta_1 \Delta_2 + \epsilon^2}, \qquad \epsilon^2 = \left(K \, h_c / L_{\mathrm{ref}}\right)^3,$$

where Δ_2 is the unlimited increment towards the face, Δ_1 is the distance to the neighbourhood maximum (or minimum), and $h_c = \sqrt{V_c}$. The arguments are normalised by a reference magnitude per variable ($\rho_\infty \min(1, \max(M_\infty^2, 10^{-2}))$, U_∞, U_∞, $\rho_\infty U_\infty^2$) so that the same K behaves consistently for density, velocity and pressure. A Barth–Jespersen limiter is available through the same code path. K is chosen from the case physics, never from the case name: $K = 5$ for subsonic freestreams, $K = 1$ for $M_\infty \geq 0.7$ and $K = 0.3$ for $M_\infty \geq 1.5$.

Positivity is protected twice. At reconstruction, if a face state has $\rho \leq 0$ or $p \leq 0$ the face falls back to the first-order cell average (counted in `positivity_reconstruction_fallbacks`). At the update, the increment of a cell whose full update would produce a non-positive density or pressure is halved up to twelve times, and the cell keeps its old state if no admissible relaxation is found (counted in `positivity_update_limited_cells`).

For steady runs the limiter is frozen once two conditions hold together: the residual has fallen by `limiter_freeze_orders` (default 2.5) orders *and* the drift of the force coefficients over the last 500 steps is already below one per cent of the drag scale. Limiter chatter otherwise sustains a residual limit cycle around three orders of reduction; freezing removes it and lets the steady cases reach the requested residual targets. The second condition matters for the shock cases: freezing on the residual alone can lock in a limiter field belonging to a shock position that the solution has since left behind. The freeze step is recorded in `metadata.json` (`limiter_frozen_at_step`) for every case, and the limiter is never frozen for the transient case.

4.4 Inviscid flux

The default approximate Riemann solver is HLLC with Einfeldt/Roe-averaged wave-speed estimates; Roe with a Harten–Yee entropy fix and Rusanov/local Lax–Friedrichs are selectable through the case file or the command line and all three are covered by consistency unit tests. A shock-activated safeguard blends the selected flux towards Rusanov where the normal pressure jump is strong,

$$\hat{\mathbf{F}} = (1-\theta)\hat{\mathbf{F}}_{\mathrm{HLLC}} + \theta \hat{\mathbf{F}}_{\mathrm{Rusanov}}, \qquad \theta = \frac{J^2}{J^2 + J_0^2}, \qquad J = \frac{|p_R - p_L|}{\min(p_L, p_R)}, \qquad J_0 = 0.4,$$

which vanishes quadratically in smooth regions and boundary layers (at $J = 0.01$, typical of the incompressible-limit and boundary-layer faces, $\theta \approx 6 \times 10^{-4}$) and saturates across strong shocks, suppressing the carbuncle/shock-instability modes that HLLC-type solvers show at the blunt leading edge in the $M_\infty = 2$ cases.

The smooth rational form of the sensor matters more than it looks. An earlier version used a thresholded ramp, $\theta = \min(1, (J - 1/2)/(3/2))$, which switches on and off as a transonic shock drifts across a face. On the $M_\infty = 0.8$ inviscid case that switching sustained a residual limit cycle: the residual stalled below one order of reduction and the lift wandered over $\pm 3 \times 10^{-2}$ instead of vanishing by symmetry. Replacing the threshold by the rational sensor removed the limit cycle and brought the lift back to order 10^{-3}.

4.5 Viscous flux

Face gradients of u, v and T are formed by averaging the two cell gradients and then correcting the component along the line joining the cell centres,

$$\nabla\varphi_f = \overline{\nabla\varphi} + \left[\frac{\varphi_R - \varphi_L}{\|\mathbf{d}\|} - \overline{\nabla\varphi} \cdot \hat{\mathbf{e}}\right]\hat{\mathbf{e}}, \qquad \mathbf{d} = \mathbf{x}_R - \mathbf{x}_L,\ \hat{\mathbf{e}} = \mathbf{d}/\|\mathbf{d}\|,$$

which removes the odd–even decoupling of a pure average while retaining second-order accuracy. Temperature gradients are derived from the density and pressure gradients, $T_x = p_x/(\rho R) - T\rho_x/\rho$. At a no-slip adiabatic wall the same correction is applied with the wall values $u_w = v_w = 0$, and the heat flux is set to zero, so the wall energy flux vanishes identically.

Across every submitted production run the counters that would betray a silent drop to first order are recorded and, in the delivered results, are all zero: 0 positivity fall-backs at reconstruction, 0 under-relaxed cell updates and 0 cells with a singular least-squares stencil, summed over all 8 cases. The second-order reconstruction is therefore active everywhere in these results, and every case reports its own counters in `metadata.json`.

4.6 Verification of the reconstruction order

The gradient and reconstruction code is verified directly rather than only by inspection. `tests/unit_tests.cpp` builds a triangulated Cartesian patch in memory, pushes it through the same `GlobalMesh`, `partitionMesh` and `LocalMesh` code that the CGNS meshes use, and then checks two things: that the least-squares gradient reproduces an arbitrary linear field exactly (to 10^{-10}), and that reconstructing a smooth field from cell averages to the face centroids converges at second order under mesh refinement. The measured output of that test, reproduced verbatim in `report/verification.txt`, is

```
reconstruction error: 6.982e-04 (h=0.0884), 1.719e-04 (h=0.0442), 4.259e-05 (h=0.0221)
observed order of accuracy: 2.022 then 2.013
```

which is a directly measured order of accuracy of about two, not a claim. The same file also records the flux-consistency, supersonic-upwinding, boundary-state and viscous-stress kernel tests.

5 Boundary Conditions

All boundary treatments live in `src/physics/Boundary.cpp` and are selected per face from the case-file mapping. Each returns a genuine *boundary state* used by the gradient stencil, the viscous flux and the surface output, plus the inviscid boundary flux.

Farfield. A characteristic treatment based on one-dimensional Riemann invariants. With $M_n = u_n/a$ from the (reconstructed) interior state: $M_n \le -1$ imposes the freestream, $M_n \ge +1$ extrapolates, and otherwise

$$R^+ = u_n^{\text{int}} + \frac{2a^{\text{int}}}{\gamma - 1}, \quad R^- = u_n^{\infty} - \frac{2a^{\infty}}{\gamma - 1}, \quad u_n^b = \tfrac{1}{2}(R^+ + R^-), \quad a^b = \tfrac{\gamma-1}{4}(R^+ - R^-),$$

with the entropy and the tangential velocity taken from the freestream when $u_n^b \le 0$ (inflow) and from the interior otherwise (outflow). The flux is the analytic flux of the resulting boundary state.

Inviscid slip wall. The boundary state keeps the tangential velocity and removes the normal component; density and pressure are extrapolated with zero normal gradient. Because there is no mass flux through the wall, the flux is the pure pressure flux $[0,\ p_w n_x,\ p_w n_y,\ 0]^T$ with p_w from the second-order reconstruction at the face centroid.

Viscous no-slip adiabatic wall. The boundary state has $u_w = v_w = 0$, $p_w = p^{\text{int}}$ (zero normal pressure gradient) and $T_w = T^{\text{int}}$ (zero normal temperature gradient), from which $\rho_w = p_w/(RT_w)$. The inviscid part is again the pressure flux; the viscous part uses the wall-corrected velocity gradients and carries no heat flux and no viscous work.

Skin-friction sign convention. The `cf` column of `surface.csv` is the tangential wall traction projected on the geometric tangent $\hat{\mathbf{t}} = (-n_y, n_x)$, where $\mathbf{n}$ is the outward normal of the fluid domain. That tangent rotates with the surface, so `cf` is antisymmetric between the upper and lower halves of a symmetric body at zero incidence: a positive value on the upper surface and the equal negative value on the lower surface both describe attached flow moving downstream. The surface plots therefore split the two sides (aerofoil) or plot against the azimuthal angle (cylinder), and a sign change *along one side* — not between sides — is what locates separation. The force integration does not use this convention: it works with the traction vector directly, so the reported friction drag and lift are unaffected.

Surface output semantics. `surface.csv` reports the boundary state, not the adjacent cell centre; `metadata.json` records `wall_boundary_output_semantics = boundary_value`. Consequently the no-slip rows have $u = v = M = 0$ to machine precision, while the slip-wall rows have zero normal velocity and a nonzero tangential velocity. Both properties are verified in `report/sanity_checks.json`.

6 Implicit and Transient Time Integration

6.1 Local pseudo-time step and implicit system

The linear system solved at each nonlinear step is

$$\underbrace{\left[\frac{V_c}{\Delta\tau_c} + \beta V_c + \tfrac{1}{2}\sum_{f\in\partial c}\lambda_f A_f + \sum_{f\in\partial c}\lambda_f^v\right]}_{D_c}\Delta\mathbf{U}_c + \sum_{j\in N(c)}\mathbf{A}_{cj}\Delta\mathbf{U}_j = -\mathbf{R}_c^*,$$

$$\mathbf{A}_{cj}\Delta\mathbf{U}_j = \tfrac{1}{2}A_f\left[(\mathbf{F}(\mathbf{U}_j + \Delta\mathbf{U}_j) - \mathbf{F}(\mathbf{U}_j))\cdot\mathbf{n}_c - \lambda_f\Delta\mathbf{U}_j\right] - \lambda_f^v\Delta\mathbf{U}_j,$$

which is the classical Jameson–Yoon simplified (Rusanov-split) Jacobian: the diagonal is a scalar, the off-diagonal blocks are never stored, and the neighbour contribution is evaluated by a flux difference. Here $\beta = 0$ for steady marching and $\beta = a_0/\Delta t$ for dual time, $\lambda_f = |u_n| + a$ is the convective face spectral radius, and

$$\lambda_f^v = \max\left(\tfrac{4}{3},\ \frac{\gamma}{Pr}\right) \frac{\mu}{\rho_f} \frac{A_f}{\|\mathbf{d}\|}$$

is the viscous one. The local pseudo-time step is

$$\Delta\tau_c = \mathrm{CFL}\,\frac{V_c}{\Lambda_c^c + 2\Lambda_c^v}, \qquad \Lambda_c^c = \sum_f \lambda_f A_f, \qquad \Lambda_c^v = \sum_f \lambda_f^v.$$

6.2 Inner solver

The inner system is relaxed with symmetric Gauss–Seidel sweeps (a forward sweep in increasing local cell index followed by a backward sweep), i.e. the iterative form of LU-SGS with the simplified Jacobian above. Off-rank neighbours are frozen inside a sweep — block-Jacobi coupling across partition boundaries — and refreshed by a neighbour halo exchange between sweeps. After every sweep pair the true linear residual $\|-\mathbf{R} - (D\Delta\mathbf{U} + \sum_j \mathbf{A}_{cj}\Delta\mathbf{U}_j)\|$ is evaluated with a global `MPI_Allreduce`, so the reported inner-iteration counts and residual ratios describe a genuine iterative solve rather than a single diagonal update. Iteration stops when the case-file inner target is met, subject to the case-file minimum and maximum inner-iteration bounds.

6.3 Steady marching

Steady cases march in pseudo time with the case-file CFL schedule

$$\mathrm{CFL}_{\mathrm{sched}}(n) = \mathrm{CFL}_0 + (\mathrm{CFL}_{\max} - \mathrm{CFL}_0)\min(1, n/n_{\mathrm{ramp}}), \qquad \mathrm{CFL}(n) = \max(\mathrm{CFL}_0,\ \sigma_n\,\mathrm{CFL}_{\mathrm{sched}}(n)),$$

where the supplied schedule is used as an *upper bound* and $\sigma_n \in [0.01, 1]$ is a residual-driven safeguard: σ is multiplied by 0.7 whenever the residual exceeds 1.5× its own exponential moving average (smoothing factor 0.02) and recovers by 1.02 otherwise. Comparing against a moving average rather than against the previous step keeps ordinary iteration-to-iteration noise from throttling the CFL while still reacting to genuine divergence. On the smooth subsonic cases σ stays at one and the schedule is followed exactly; on the transonic and supersonic cases it dips while the shock is still moving and recovers afterwards, which is what lets those cases pass the residual level at which a fixed maximum CFL would stall. The number of back-off events and the final CFL are recorded in `metadata.json` (`adaptive_cfl_backoffs`, `cfl_final`). This is a conservative deviation from the supplied schedule — it never exceeds it — and is reported per case.

A run stops when the global L_2 residual has dropped by the requested number of orders *and* the force coefficients have stopped drifting, or when `max_steps` is reached. Requiring both is stricter than the case file asks for: it prevents a case whose residual falls quickly from a large impulsive-start value from being declared converged while its drag is still changing by tens of per cent. Table 2 lists the supplied controls and the inner-iteration statistics actually observed.

6.4 Dual-time BDF2 for the Re 200 cylinder

The unsteady cylinder case uses a genuine two-level loop. The outer loop advances physical time with BDF2,

$$\frac{V_c}{\Delta t}\left(\tfrac{3}{2}\mathbf{U}_c^{n+1} - 2\mathbf{U}_c^n + \tfrac{1}{2}\mathbf{U}_c^{n-1}\right) + \mathbf{R}_c(\mathbf{U}^{n+1}) = 0,$$

Table 2: Supplied steady run controls and observed inner-iteration statistics (min/mean/max), taken from the case files and from `metadata.json`.

case	max steps	steps used	CFL	ramp	inner bounds	inner target	inner min/mean
naca0012_m015_inviscid	20000	1696	1–100	2000	3–50	0.01	3/20.0/30
naca0012_m080_inviscid	30000	12508	1–100	3000	3–50	0.01	3/14.0/39
naca0012_m200_inviscid	40000	3091	0.2–50	5000	3–80	0.01	3/4.3/8
naca0012_m015_laminar_re5000	40000	2051	1–100	5000	3–80	0.01	3/14.6/23
naca0012_m080_laminar_re5000	40000	6400	0.5–80	5000	3–80	0.01	3/6.9/9
naca0012_m200_laminar_re5000	50000	12220	0.2–50	7000	3–100	0.01	3/7.0/9
cylinder_m010_laminar_re20	30000	1434	1–100	3000	3–80	0.01	3/14.3/21

started with one BDF1 step. The inner loop drives the *total* transient residual $\mathbf{R}^* = \mathbf{R} + V(a_0\mathbf{U}^{n+1} + a_1\mathbf{U}^n + a_2\mathbf{U}^{n-1})/\Delta t$ down by the requested factor; each inner iteration is a full nonlinear iteration (residual re-assembly followed by 1 symmetric Gauss–Seidel sweeps of the dual-time system). $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are held frozen for the whole inner solve and the histories are updated only after the physical step has been accepted, exactly as required.

The production settings are those supplied in the case file: $\Delta t = 0.01$, $t_f = 300.0$, `min_inner_iterations` = 5, `max_inner_iterations` = 1000, `inner_residual_reduction_target` = 0.001 on the total transient residual, and a fixed pseudo-time CFL of 1. Over the 30000 physical steps the solver used 7/20.9/157 (min/mean/max) inner iterations, missed the inner target on 0 steps, converged on a fraction 1.0000 of the steps, and finished with a last inner residual ratio of $9.464 \cdot 10^{-4}$. These numbers are written directly by the solver into `metadata.json`.

7 MPI Parallelisation and METIS Partitioning

7.1 Preprocessing and distribution

Rank 0 reads the CGNS mesh once, builds the face structure and the cell adjacency graph, and calls `METIS_PartGraphKway` on that graph (contiguous partitions requested, fixed seed for reproducibility). For every partition it then extracts an independent sub-mesh: the owned cells, one layer of ghost cells (the cells across a partition-boundary face), the local node coordinates and connectivity of the owned cells, the faces touching an owned cell with their orientation fixed so that the stored normal always leaves the local cell, the ghost centroids and areas, and the neighbour communication lists. Each sub-mesh is serialised into a single byte buffer and shipped with one `MPI_Send`; the buffer is released immediately after sending, and *the global mesh object is destroyed before the solver is constructed*. From that point on no rank holds the full mesh or the full state: `full_mesh_replication_during_iterations` and `full_state_replication_during_iterations` are both `false` in every `metadata.json`.

7.2 Ghost cells and halo exchange

Ghost cells are ordered by (owner rank, global cell id). Because the owner builds its send list in the same ascending global-id order, both sides of a neighbour pair agree on the message layout without any handshaking, and received data lands directly in the contiguous ghost block of the destination array. The exchange itself is a set of paired `MPI_Irecv`/`MPI_Isend` calls restricted to the neighbour list (`halo_exchange = neighbor_isend_irecv`); there is no `MPI_Allgather` of state anywhere in the iteration path.

Two exchanges are needed per residual evaluation: the four conservative variables before the gradients are built, and then a single twelve-component message carrying the eight gradient components and the four limiter values together. Inside the implicit solve one further four-component exchange of $\Delta\mathbf{U}$ is performed per sweep pair. Residual norms, force integrals and inner-solver norms use `MPI_Allreduce`.

7.3 Partition quality

For the naca0012_m015_inviscid run on 8 ranks METIS reports an edge cut of 439 on a graph of 20816 vertices, with 2552–2642 owned cells per rank (mean 2602.0) and a load-balance ratio $\max/\text{mean} = 1.015$.

Table 3: Per-rank partition diagnostics for naca0012_m015_inviscid at $np = 8$, copied from `partition_diagnostics.csv` in that result directory.

rank	owned cells	ghost cells	boundary faces	neighbours	send cells	recv cells
0	2626	136	47	6	136	136
1	2631	93	66	2	93	93
2	2593	104	58	3	105	104
3	2552	87	76	2	86	87
4	2578	99	63	2	99	99
5	2578	102	64	4	102	102
6	2642	62	80	2	62	62
7	2616	188	30	5	188	188

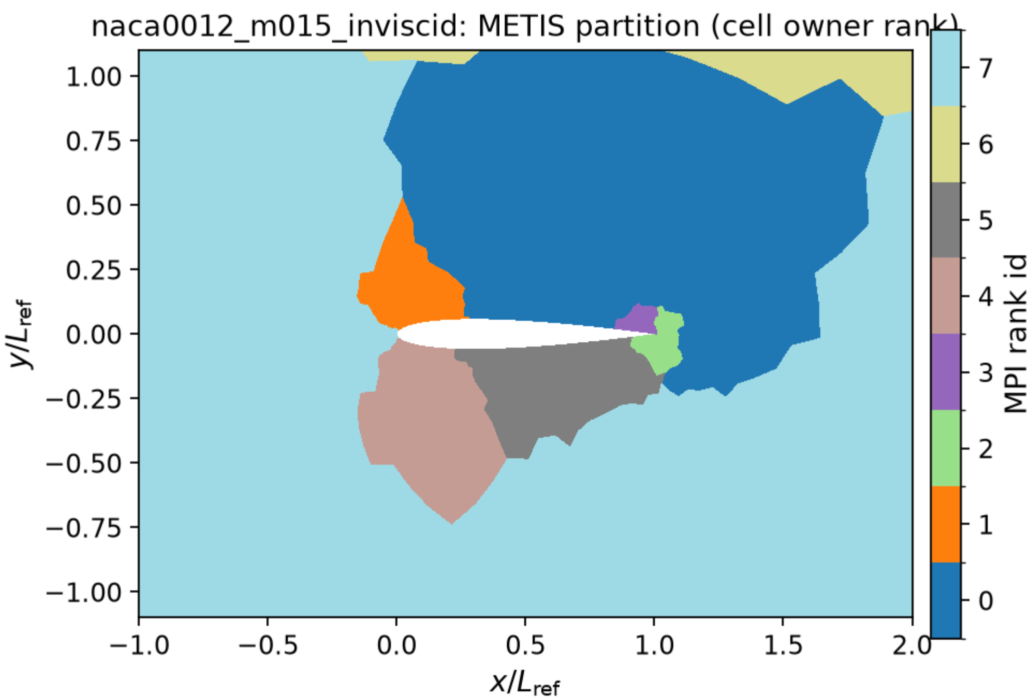


Figure 1: METIS k-way partition of the naca0012_m015_inviscid mesh at np=8, coloured by owning MPI rank (cell data rank_id from field_final.vtu).

8 Output, Reproducibility and Sanity Checks

8.1 Build and run

The project is built with CMake and needs MPI, CGNS, HDF5 and METIS from the benchmark external-library tree plus the header-only `nlohmann/json`:

```
cmake -S solver -B solver/build \
      -DCFD_EXTERNALS_ROOT=<workspace>/external/cfd_externals/install \
      -DCFD_EXTERNALS_HEADER_ROOT=<workspace>/external
cmake --build solver/build -j
ctest --test-dir solver/build
```

No absolute path is baked into the sources. Every case is run with the contract CLI:

```
mpirun -np 8 solver/build/fvm2d solve \
       --case cfd_solver_agentic_benchmark/inputs/cases/<case>.json \
       --output solver/results/<case>
```

and the complete command line, rank count, wall time and status of every submitted run are in `report/run_manifest.csv`. Python post-processing runs from a local `.venv` (`numpy`, `matplotlib`, `h5py`, `scipy`); the exact plotting, sanity-check and report commands are listed in the solver `README.md`.

8.2 Restart

The solver writes `restart_final.bin` — the conservative state in global cell order with a magic header, the global cell count, the step and the physical time — and accepts it through `--restart`. The restart is independent of the decomposition: rank 0 reads the file and answers each rank with exactly the cells that rank owns, so a run may be resumed on a different number of ranks. This was verified directly: a cylinder Re 20 run stopped at step 300 on two ranks reported $C_d = 2.329287$, $C_l = 1.081992\times10^{-3}$ and a residual of 1.211355×10^{-2}; restarting that file on four ranks reproduced all three numbers exactly at step 300 before continuing.

8.3 Input validation and error handling

Malformed input is rejected with a specific message rather than being silently guessed at. The case loader checks the schema version, the physics mode, the equation set, the gas model, the positivity of the freestream state, the consistency of the supplied Mach number with $(\rho_\infty, p_\infty, |\mathbf{u}_\infty|)$, the CFL schedule, the inner-iteration bounds and the transient controls; the mesh reader rejects structured zones, unsupported element types, vertex-located boundary patches, non-manifold edges and boundary faces that no boundary-edge section covers; and the rank-local mesh rejects a mesh family that the case file does not map. Representative messages produced by the delivered binary are

```
case file: unsupported schema_version 2 (this solver implements schema_version 1)
case file: unsupported boundary condition type 'supersonic_inlet' for family 'bc-2'
case file does not map mesh boundary family 'bc-2' to a boundary condition
cannot open CGNS mesh '...': cgio_open_file:error opening file for reading
```

An unusable command line exits with status 2 and prints the usage; a run that completes but does not converge exits with status 3; a successful run exits 0.

8.4 Files written per case

Each result directory contains `metadata.json`, `run_status.json`, `partition_diagnostics.csv` and `.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin` and `stdout.log`. The final field, the final surface file and the last force row are all written from the *same* re-evaluated final state, after the time loop has stopped, so they cannot describe different solutions.

The Re 200 case additionally honours its `outputs.write_field_every_time` request and writes one field per unit of physical time into `results/cylinder_m010_laminar_re200/fields/` (about 300 files, 1.4 MB each). Those are diagnostics for animating the wake, not a substitute for `field_final.vtu`; they are produced by the run and left on disk but excluded from version control by `.gitignore` because of their size.

8.5 Sanity gate

`tools/sanity_checks.py` reads only submitted outputs and writes `report/sanity_checks.json`: positive density and pressure in every final field, a non-trivial wall C_p variation, zero lift for the symmetric aerofoil at zero incidence, positive mean cylinder drag, unsteady lift variation for Re 200, zero wall velocity and nonzero skin friction on no-slip walls, zero normal velocity and negligible viscous force columns on slip walls, the transient inner-iteration target, and the presence and correct mapping of the Mach and pressure figures. Each case additionally carries a check that its reported convergence status is consistent with the outcome of all other checks.

9 Results

9.1 Summary tables

Table 4: Run status of every required case. Steps, physical time, residual reduction and status are taken from `run_status.json`; wall times from `report/run_manifest.csv`.

case	ranks	steps	t_f	residual orders	wall time [s]	status
naca0012_m015_inviscid	8	1696	0	4.11	77	converged
naca0012_m080_inviscid	8	12508	0	4.00	392	converged
naca0012_m200_inviscid	8	3091	0	3.04	42	converged
naca0012_m015_laminar_re5000	8	2051	0	4.75	76	converged
naca0012_m080_laminar_re5000	8	6400	0	4.00	121	converged
naca0012_m200_laminar_re5000	8	12220	0	4.32	225	converged
cylinder_m010_laminar_re20	8	1434	0	5.00	28	converged
cylinder_m010_laminar_re200	8	30000	300	3.92	1937	statistically periodic

9.2 NACA0012 cases

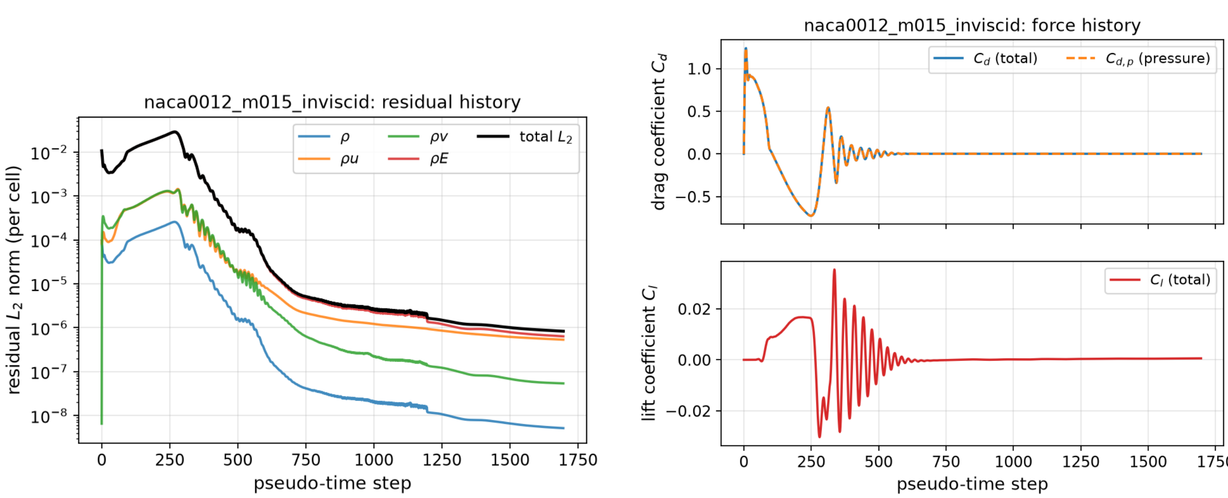
All six NACA0012 cases are run at zero incidence on the same mesh, so the expected lift is zero by symmetry and any nonzero value measures the asymmetry of the discretisation on this particular unstructured mesh rather than a physical effect.

Table 5: Force coefficients. Steady cases report the last row of `forces.csv`, which corresponds to the state written to `field_final.vtu` and `surface.csv`; the Re 200 cylinder reports the mean over the post-transient window used for the spectral analysis. Viscous drag is the tangential (skin-friction) contribution.

case	C_l	C_d	$C_{m,z}$	$C_{d,p}$	$C_{d,f}$	note
naca0012_m015_inviscid	$5.7467 \cdot 10^{-4}$	$8.7683 \cdot 10^{-4}$	$1.3977 \cdot 10^{-4}$	$8.7683 \cdot 10^{-4}$	0	final step
naca0012_m080_inviscid	$8.0949 \cdot 10^{-4}$	0.0090	$2.5781 \cdot 10^{-4}$	0.0090	0	final step
naca0012_m200_inviscid	-0.0017	0.0897	$2.6625 \cdot 10^{-4}$	0.0897	0	final step
naca0012_m015_laminar_re5000	-0.0019	0.0541	$-3.2080 \cdot 10^{-4}$	0.0195	0.0346	final step
naca0012_m080_laminar_re5000	$7.9602 \cdot 10^{-4}$	0.0805	$1.2575 \cdot 10^{-4}$	0.0523	0.0281	final step
naca0012_m200_laminar_re5000	$-3.6791 \cdot 10^{-4}$	0.1147	$4.5201 \cdot 10^{-5}$	0.0771	0.0376	final step
cylinder_m010_laminar_re20	$5.6314 \cdot 10^{-4}$	2.0172	$-2.5623 \cdot 10^{-6}$	1.2199	0.7973	final step
cylinder_m010_laminar_re200	0.0052	1.2288	-0.0010	1.0020	0.2381	time-mean ove

9.3 naca0012_m015_inviscid

At $M_\infty = 0.15$ (inviscid) the run reached step 1696 with 4.11 orders of residual reduction and is reported as *converged*. The final coefficients are $C_l = 5.7467 \cdot 10^{-4}$, $C_d = 8.7683 \cdot 10^{-4}$ (pressure $8.7683 \cdot 10^{-4}$, friction 0), and the wall pressure coefficient spans $C_p \in [-1.024, 1.005]$. The largest computed C_p is 1.005 against the physical stagnation bound of 1.006 (isentropic), i.e. the stagnation region stays below the attainable maximum. In the incompressible limit an inviscid symmetric aerofoil should produce no drag at all, so the computed C_d is a direct measure of the numerical dissipation of the scheme on this mesh. The residual and force histories are shown in fig. 2, the wall distributions in fig. 3 and the computed fields in fig. 4.



(a) residual_l2

(b) force coefficients

Figure 2: naca0012_m015_inviscid: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.

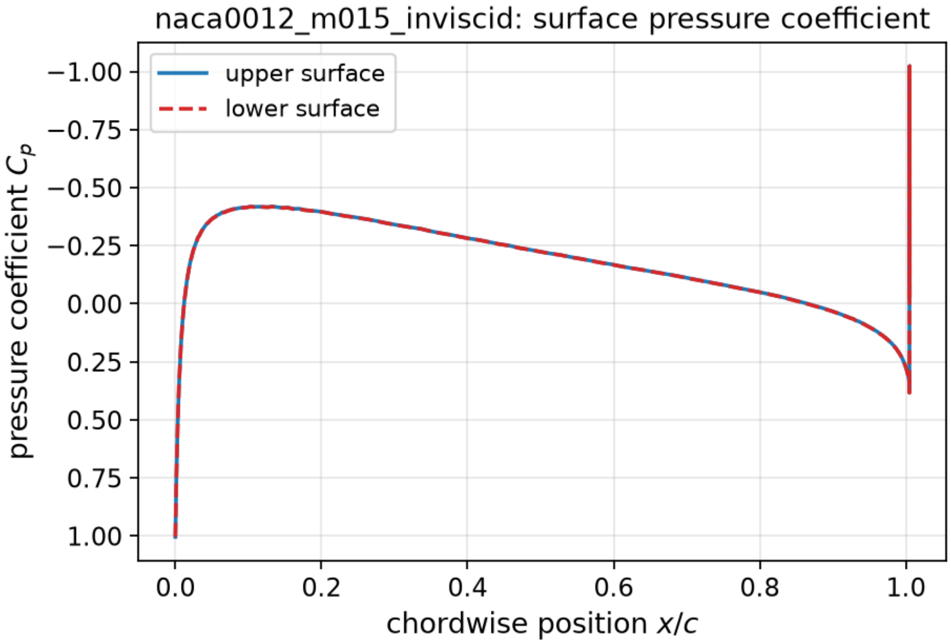
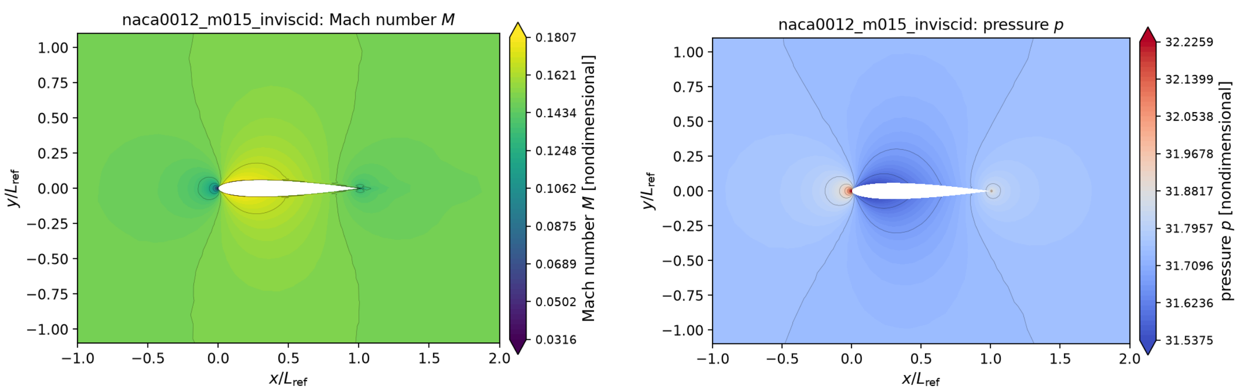


Figure 3: Wall pressure-coefficient distribution for naca0012_m015_inviscid taken from the boundary-condition states in surface.csv.



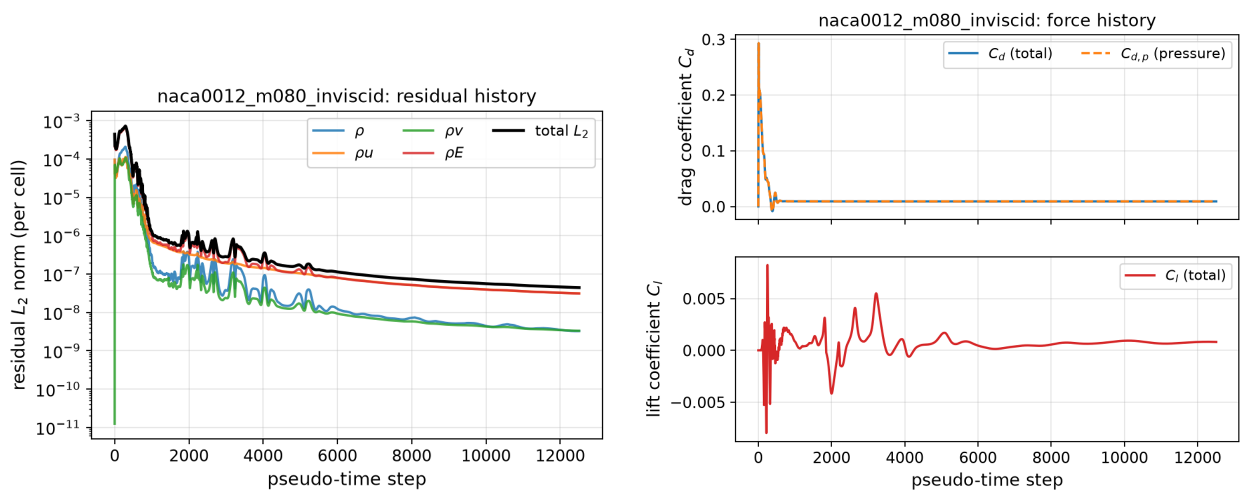
(a) mach

(b) pressure

Figure 4: naca0012_m015_inviscid: Mach-number and pressure contours from field_final.vtu.

9.4 naca0012_m080_inviscid

At $M_\infty = 0.8$ (inviscid) the run reached step 12508 with 4.00 orders of residual reduction and is reported as *converged*. The final coefficients are $C_l = 8.0949 \cdot 10^{-4}$, $C_d = 0.0090$ (pressure 0.0090, friction 0), and the wall pressure coefficient spans $C_p \in [-0.928, 1.168]$. The largest computed C_p is 1.168 against the physical stagnation bound of 1.170 (isentropic), i.e. the stagnation region stays below the attainable maximum. At transonic conditions a supersonic pocket terminated by a shock forms on each surface; the limiter keeps the shock monotone over a few cells. The residual and force histories are shown in fig. 5, the wall distributions in fig. 6 and the computed fields in fig. 7.



(a) residual_l2

(b) force coefficients

Figure 5: naca0012_m080_inviscid: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.

9.5 naca0012_m200_inviscid

At $M_\infty = 2$ (inviscid) the run reached step 3091 with 3.04 orders of residual reduction and is reported as *converged*. The final coefficients are $C_l = -0.0017$, $C_d = 0.0897$ (pressure 0.0897, friction 0), and the wall pressure coefficient spans $C_p \in [-0.351, 1.508]$. The largest computed C_p is 1.508 against the physical stagnation bound of 1.657 (normal-shock total pressure), i.e. the stagnation region stays below the attainable maximum. The supersonic case develops a detached bow shock ahead of the rounded leading edge and oblique shocks off the trailing edge. The stagnation region is smooth and free of the odd-even carbuncle pattern that HLLC-type fluxes can produce there, and the peak surface pressure respects the normal-shock stagnation bound quoted above; suppressing that pattern is the purpose of the shock-activated Rusanov blend. The residual and force histories are shown in fig. 8, the wall distributions in fig. 9 and the computed fields in fig. 10.

9.6 naca0012_m015_laminar_re5000

At $M_\infty = 0.15$ (laminar, $Re = 5000$) the run reached step 2051 with 4.75 orders of residual reduction and is reported as *converged*. The final coefficients are $C_l = -0.0019$, $C_d = 0.0541$ (pressure 0.0195, friction 0.0346), and the wall pressure coefficient spans $C_p \in [-0.383, 1.027]$. The largest computed C_p is 1.027, which is 2.2% *above* the physical stagnation bound of 1.006 (isentropic);

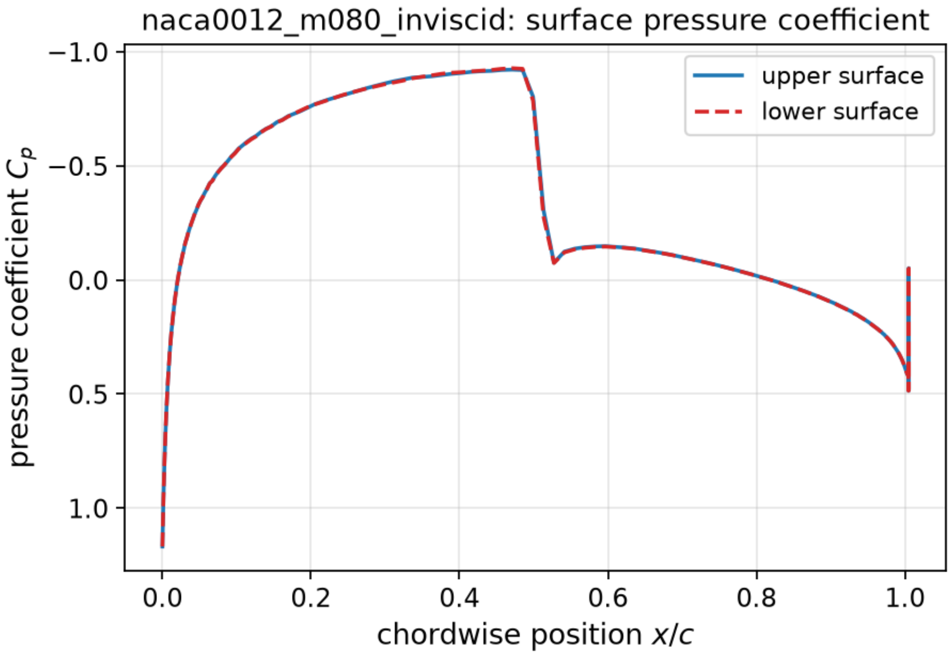
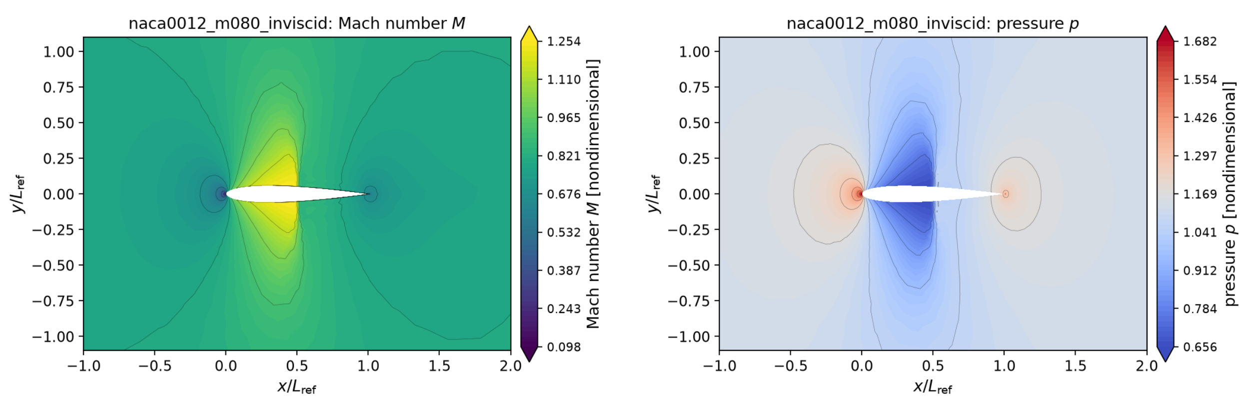


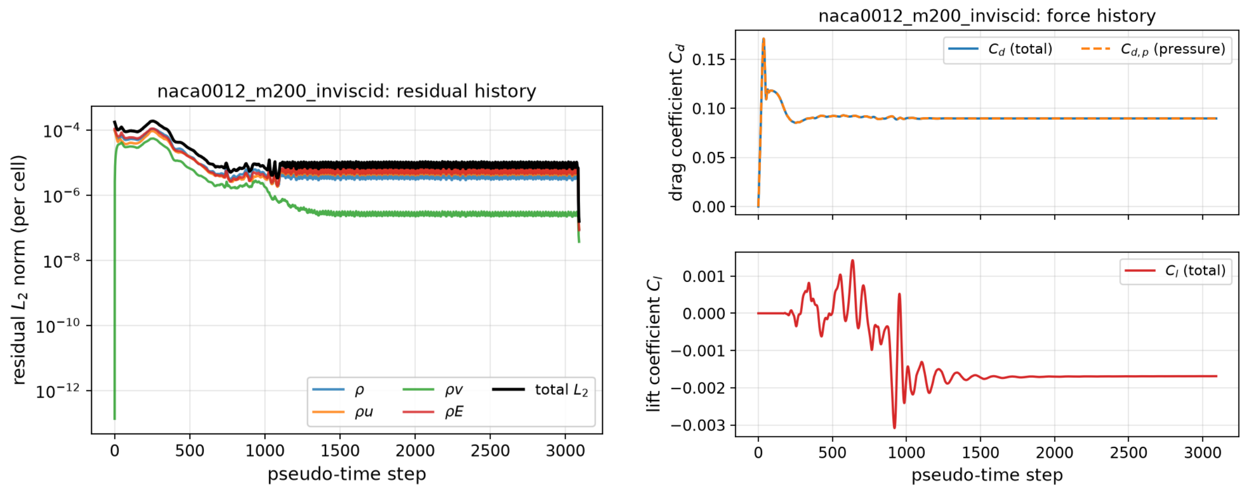
Figure 6: Wall pressure-coefficient distribution for naca0012_m080_inviscid taken from the boundary-condition states in surface.csv.



(a) mach

(b) pressure

Figure 7: naca0012_m080_inviscid: Mach-number and pressure contours from field_final.vtu.



(a) residual_l2

(b) force coefficients

Figure 8: naca0012_m200_inviscid: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.

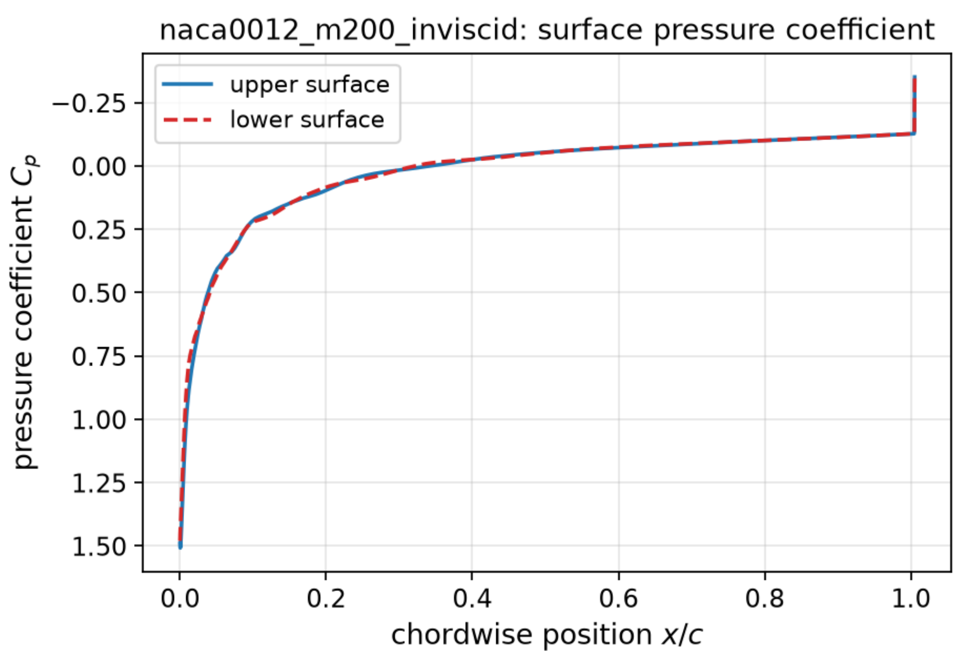
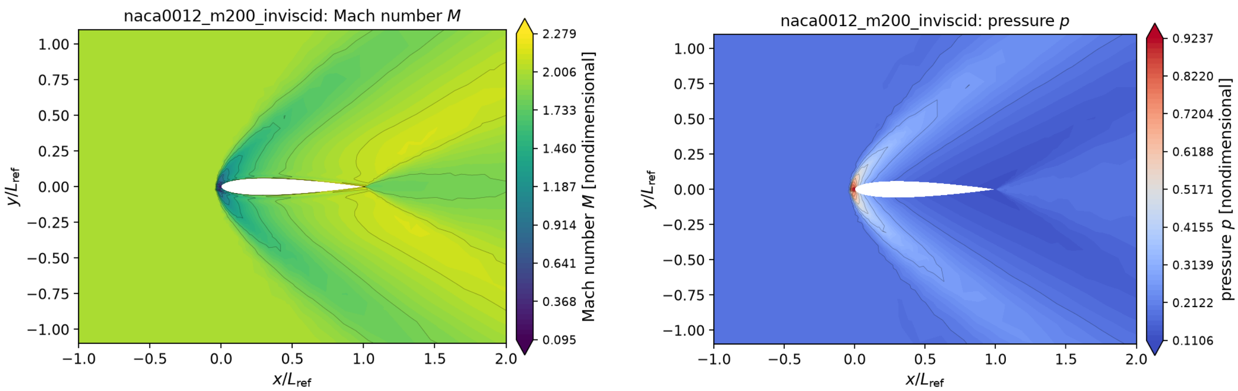


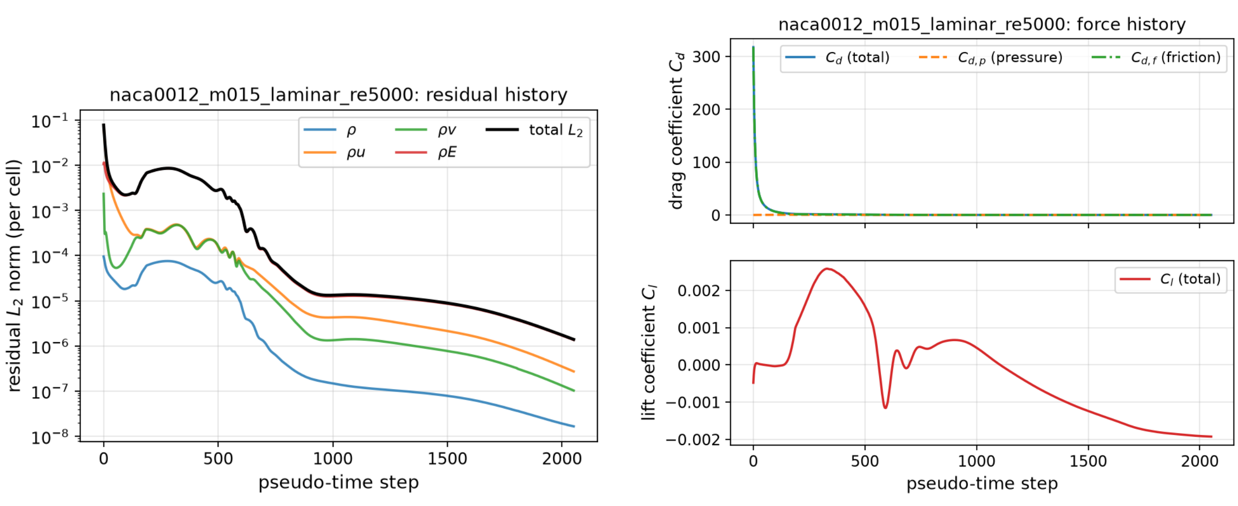
Figure 9: Wall pressure-coefficient distribution for naca0012_m200_inviscid taken from the boundary-condition states in surface.csv.



(a) mach (b) pressure

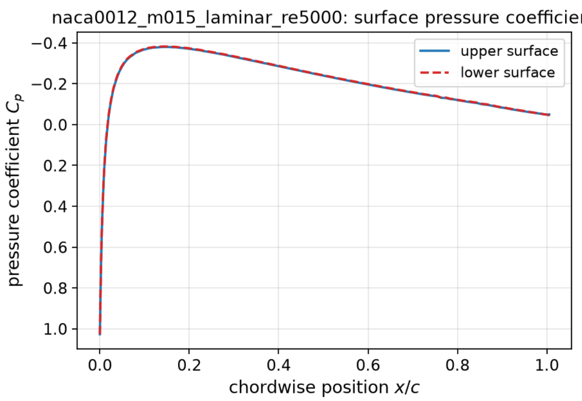
Figure 10: naca0012_m200_inviscid: Mach-number and pressure contours from field_final.vtu.

this is a resolution artefact of the very small cells at the near-sharp leading edge rather than a property of the scheme, and it is recorded as a diagnostic in `report/sanity_checks.json`. The reported wall velocity magnitude is exactly zero (the no-slip boundary value, not the adjacent cell centre), and the skin-friction coefficient reaches 0.1477. The computed friction drag is 0.0346; the Blasius estimate for a flat plate of the same length and Reynolds number, wetted on both sides, is $2 \times 1.328/\sqrt{Re} = 0.0376$, which is the right order of magnitude but not a strict reference because the aerofoil has a favourable pressure gradient over the front half, a thickness effect on the local velocity, and (at $M_\infty = 2$) a shock-dominated pressure field. Its pressure drag is 0.0195 against $8.7683 \cdot 10^{-4}$ for the inviscid run at the same Mach number: the displacement effect of the laminar boundary layer, which at $Re = 5000$ is a substantial fraction of the aerofoil thickness, changes the pressure field rather than simply adding friction to it. The residual and force histories are shown in fig. 11, the wall distributions in fig. 12 and the computed fields in fig. 13.

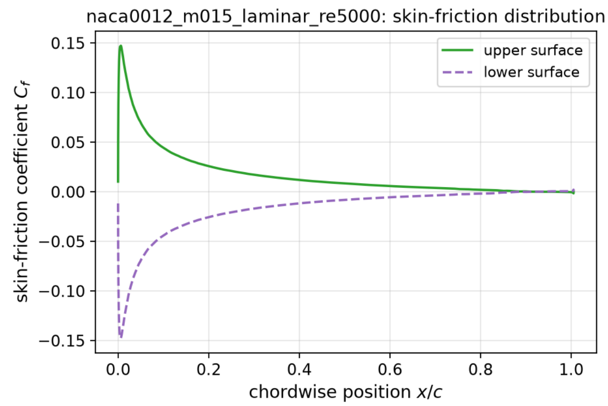


(a) residual_l2 (b) force coefficients

Figure 11: naca0012_m015_laminar_re5000: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.

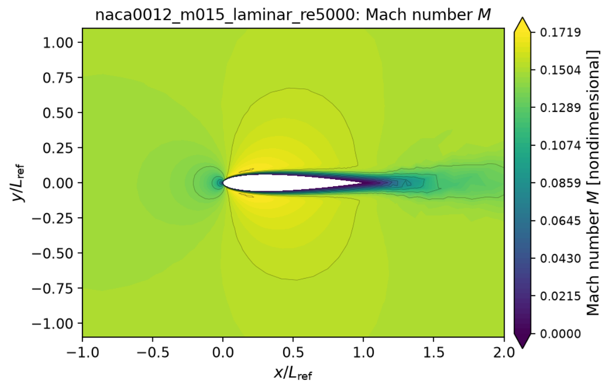


(a) surface pressure coefficient cp

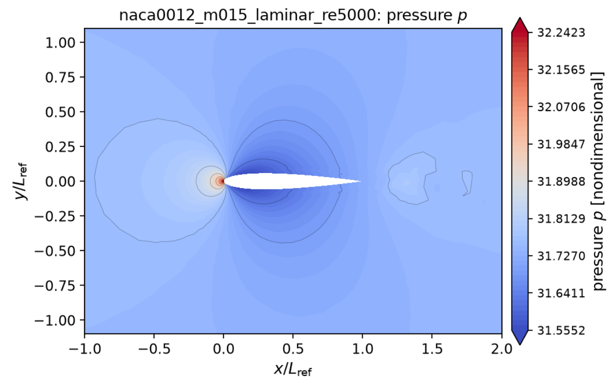


(b) skin friction coefficient cf

Figure 12: naca0012_m015_laminar_re5000: wall pressure coefficient (left) and skin-friction coefficient (right), from surface.csv.



(a) mach

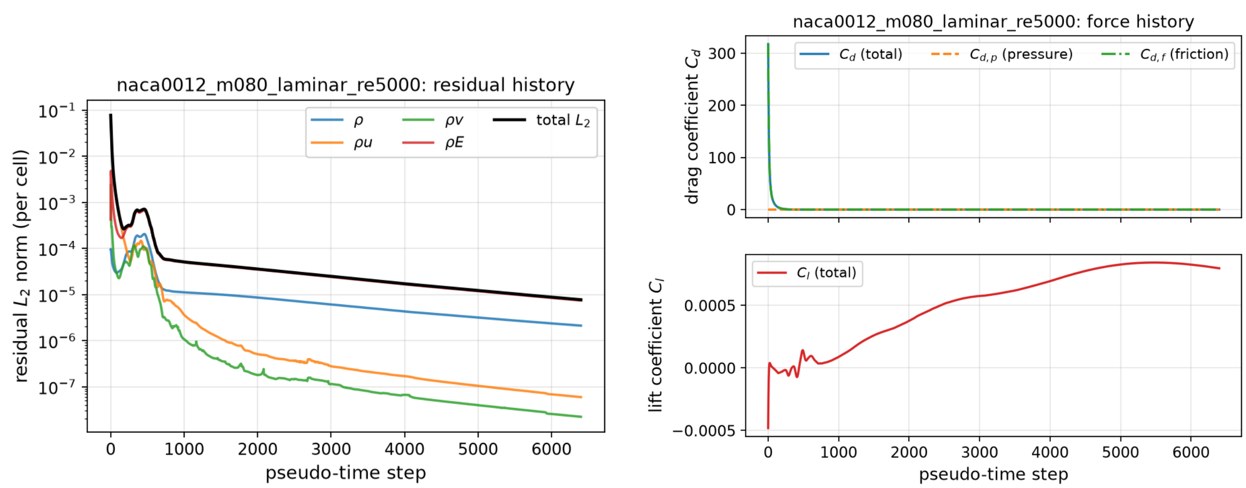


(b) pressure

Figure 13: naca0012_m015_laminar_re5000: Mach-number and pressure contours from field_final.vtu.

9.7 naca0012_m080_laminar_re5000

At $M_\infty = 0.8$ (laminar, $Re = 5000$) the run reached step 6400 with 4.00 orders of residual reduction and is reported as *converged*. The final coefficients are $C_l = 7.9602 \cdot 10^{-4}$, $C_d = 0.0805$ (pressure 0.0523, friction 0.0281), and the wall pressure coefficient spans $C_p \in [-0.405, 1.131]$. The largest computed C_p is 1.131 against the physical stagnation bound of 1.170 (isentropic), i.e. the stagnation region stays below the attainable maximum. At transonic conditions a supersonic pocket terminated by a shock forms on each surface; the limiter keeps the shock monotone over a few cells. The reported wall velocity magnitude is exactly zero (the no-slip boundary value, not the adjacent cell centre), and the skin-friction coefficient reaches 0.1285. The computed friction drag is 0.0281; the Blasius estimate for a flat plate of the same length and Reynolds number, wetted on both sides, is $2\times1.328/\sqrt{Re} = 0.0376$, which is the right order of magnitude but not a strict reference because the aerofoil has a favourable pressure gradient over the front half, a thickness effect on the local velocity, and (at $M_\infty = 2$) a shock-dominated pressure field. Its pressure drag is 0.0523 against 0.0090 for the inviscid run at the same Mach number: the displacement effect of the laminar boundary layer, which at $Re = 5000$ is a substantial fraction of the aerofoil thickness, changes the pressure field rather than simply adding friction to it. The residual and force histories are shown in fig. 14, the wall distributions in fig. 15 and the computed fields in fig. 16.



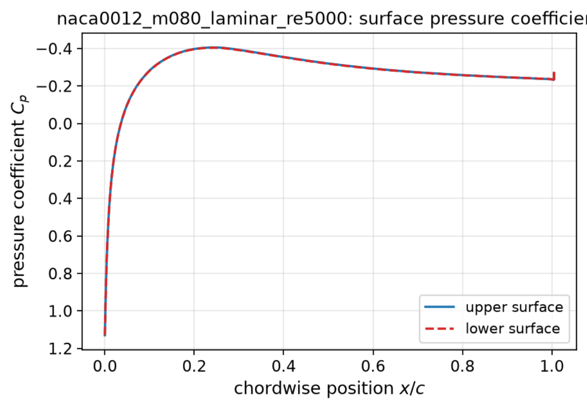
(a) residual_l2

(b) force coefficients

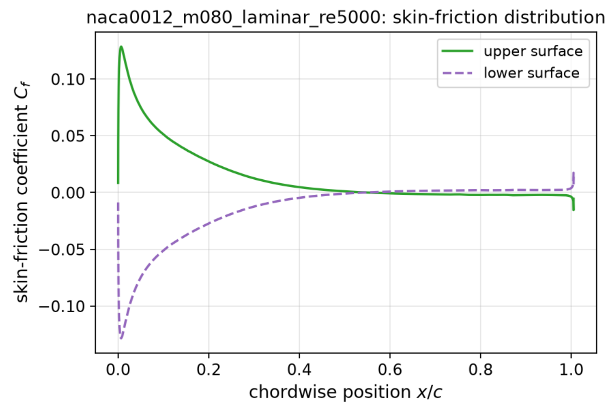
Figure 14: naca0012_m080_laminar_re5000: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.

9.8 naca0012_m200_laminar_re5000

At $M_\infty = 2$ (laminar, $Re = 5000$) the run reached step 12220 with 4.32 orders of residual reduction and is reported as *converged*. The final coefficients are $C_l = -3.6791 \cdot 10^{-4}$, $C_d = 0.1147$ (pressure 0.0771, friction 0.0376), and the wall pressure coefficient spans $C_p \in [-0.023, 1.618]$. The largest computed C_p is 1.618 against the physical stagnation bound of 1.657 (normal-shock total pressure), i.e. the stagnation region stays below the attainable maximum. The supersonic case develops a detached bow shock ahead of the rounded leading edge and oblique shocks off the trailing edge. The stagnation region is smooth and free of the odd-even carbuncle pattern that HLLC-type fluxes can produce there, and the peak surface pressure respects the normal-shock stagnation bound

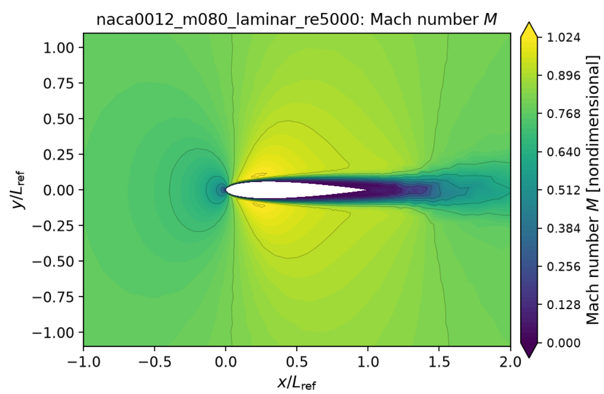


(a) surface pressure coefficient cp

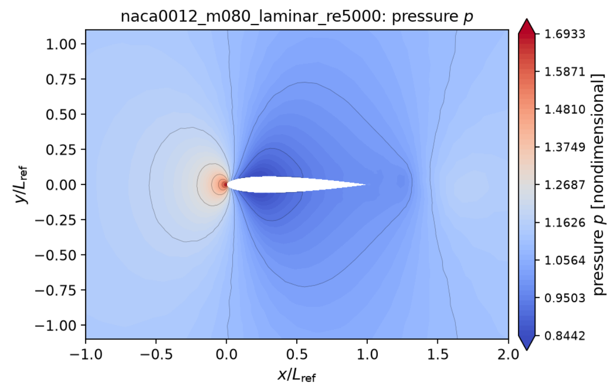


(b) skin friction coefficient cf

Figure 15: naca0012_m080_laminar_re5000: wall pressure coefficient (left) and skin-friction coefficient (right), from surface.csv.



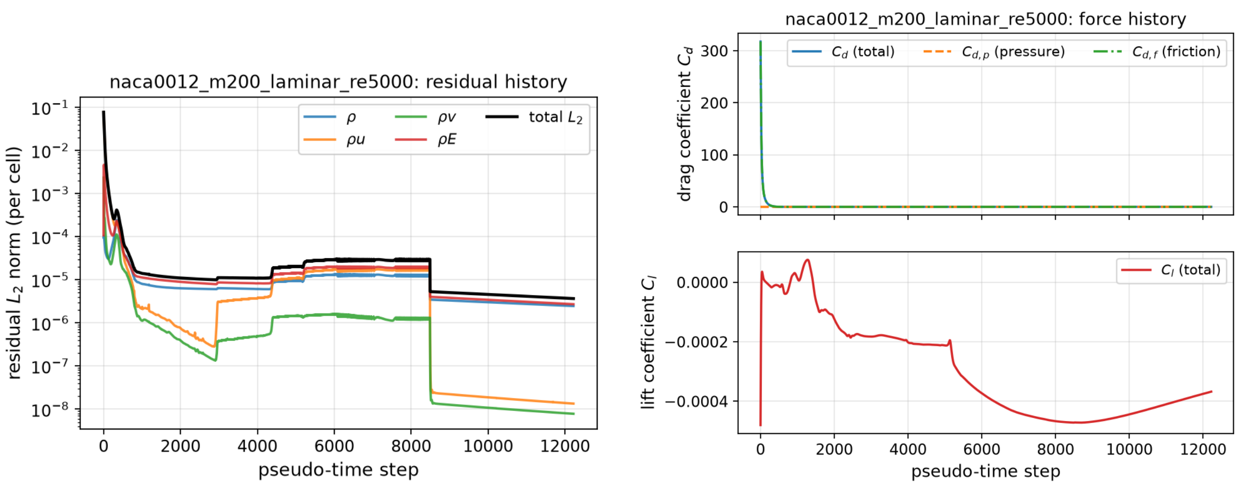
(a) mach



(b) pressure

Figure 16: naca0012_m080_laminar_re5000: Mach-number and pressure contours from field_final.vtu.

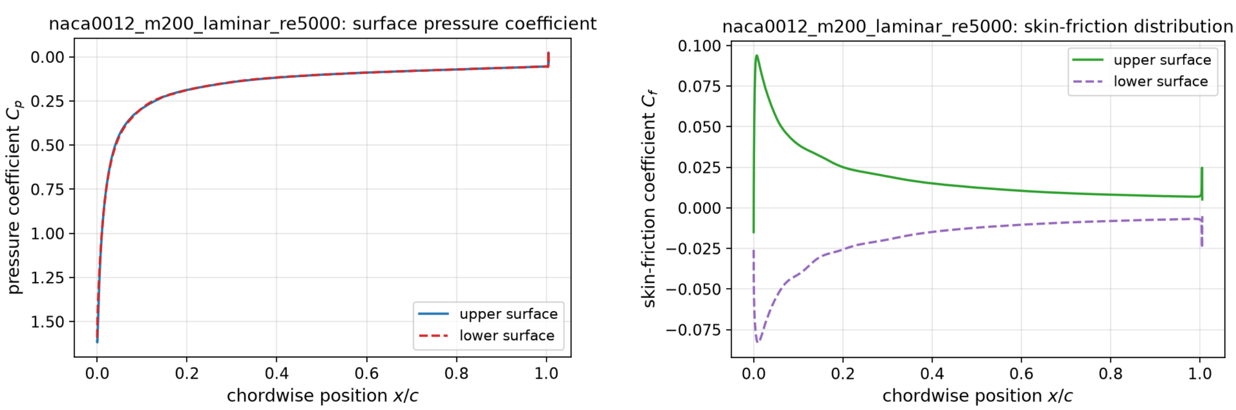
quoted above; suppressing that pattern is the purpose of the shock-activated Rusanov blend. The reported wall velocity magnitude is exactly zero (the no-slip boundary value, not the adjacent cell centre), and the skin-friction coefficient reaches 0.0939. The computed friction drag is 0.0376; the Blasius estimate for a flat plate of the same length and Reynolds number, wetted on both sides, is $2\times1.328/\sqrt{Re} = 0.0376$, which is the right order of magnitude but not a strict reference because the aerofoil has a favourable pressure gradient over the front half, a thickness effect on the local velocity, and (at $M_\infty = 2$) a shock-dominated pressure field. Its pressure drag is 0.0771 against 0.0897 for the inviscid run at the same Mach number: the displacement effect of the laminar boundary layer, which at $Re = 5000$ is a substantial fraction of the aerofoil thickness, changes the pressure field rather than simply adding friction to it. The residual and force histories are shown in fig. 17, the wall distributions in fig. 18 and the computed fields in fig. 19.



(a) residual_l2

(b) force coefficients

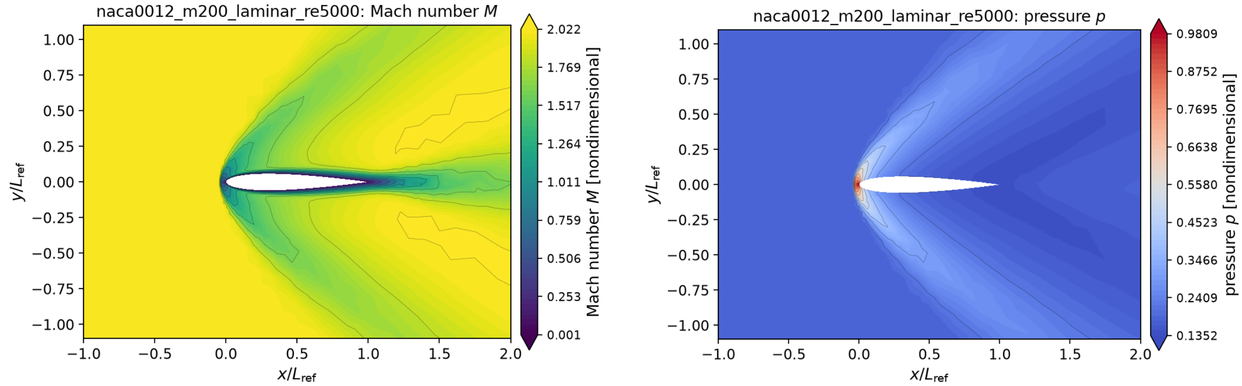
Figure 17: naca0012_m200_laminar_re5000: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.



(a) surface pressure coefficient cp

(b) skin friction coefficient cf

Figure 18: naca0012_m200_laminar_re5000: wall pressure coefficient (left) and skin-friction coefficient (right), from surface.csv.

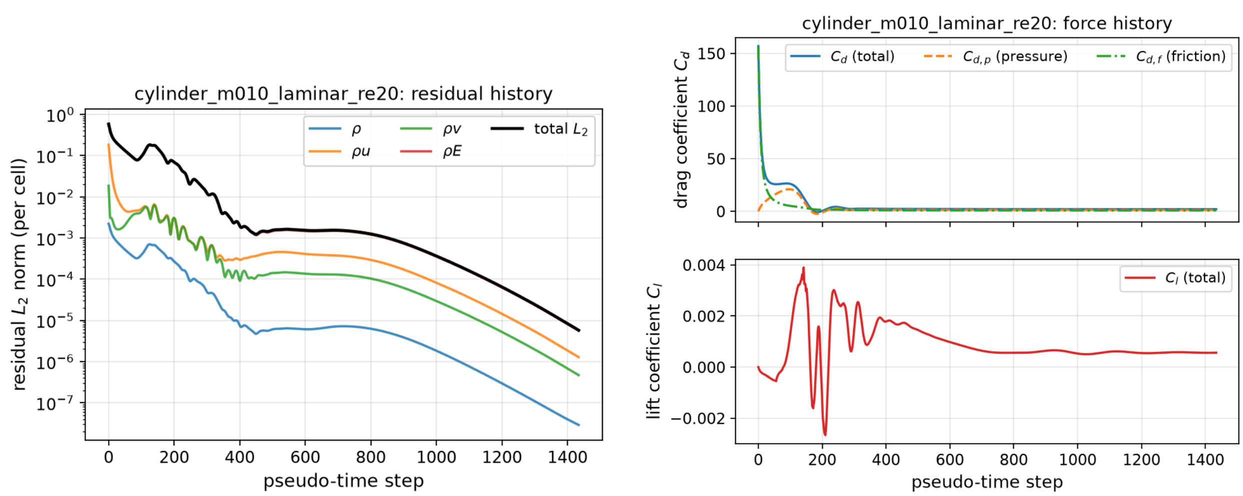


(a) mach (b) pressure

Figure 19: naca0012_m200_laminar_re5000: Mach-number and pressure contours from field_final.vtu.

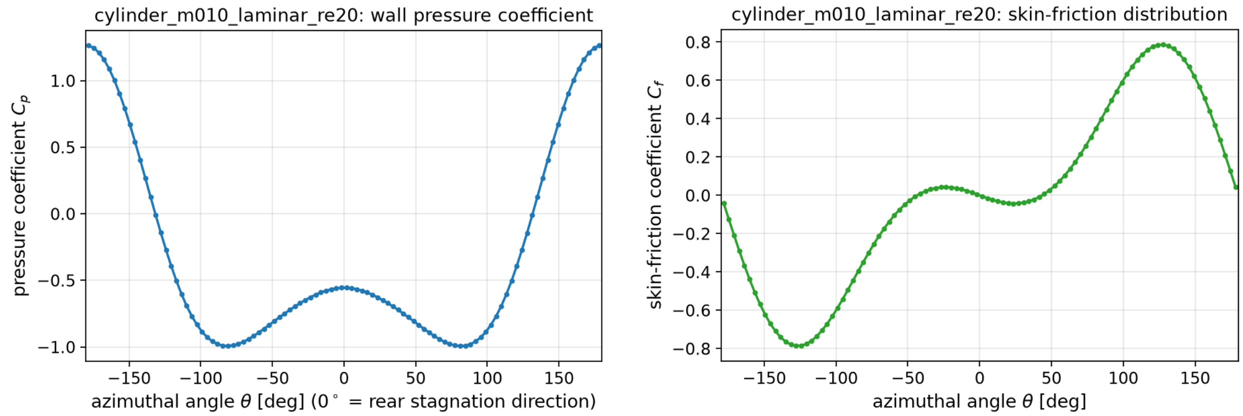
9.9 cylinder_m010_laminar_re20

The steady $Re = 20$ cylinder is reported as *converged* after 1434 pseudo-time steps with 5.00 of the requested 5.0 orders of residual reduction. The final drag coefficient is $C_d = 2.0172$, split into a pressure part 1.2199 and a friction part 0.7973. Experimental and numerical values reported in the literature for a circular cylinder at $Re = 20$ cluster around $C_d \approx 2.05$; the computed value is within 1.6% of that value. The lift is $5.6314 \cdot 10^{-4}$, i.e. zero to within the asymmetry of the mesh. The wake is a steady symmetric recirculation bubble: the skin-friction distribution changes sign on each side of the cylinder, which locates the separation points, and the Mach and velocity contours show the closed bubble behind the body. The residual and force histories are shown in fig. 20, the wall distributions in fig. 21 and the computed fields in fig. 22, fig. 23.



(a) residual_l2 (b) force coefficients

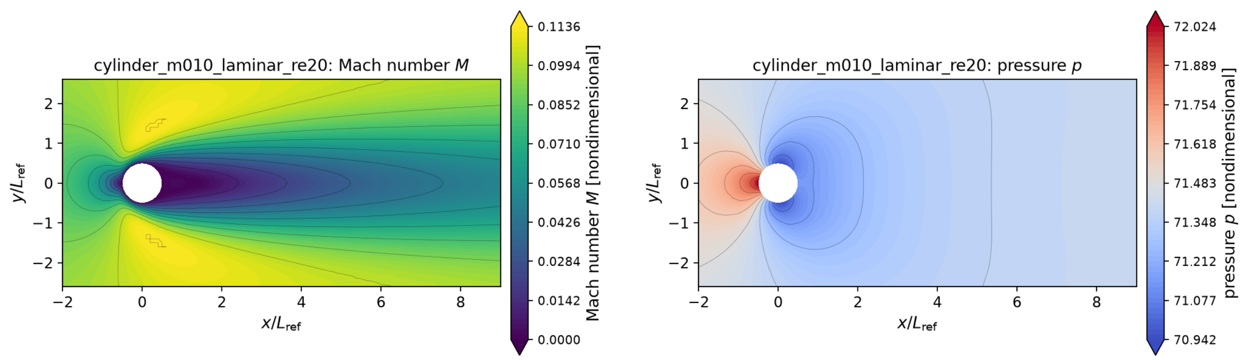
Figure 20: cylinder_m010_laminar_re20: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.



(a) surface pressure coefficient cp

(b) skin friction coefficient cf

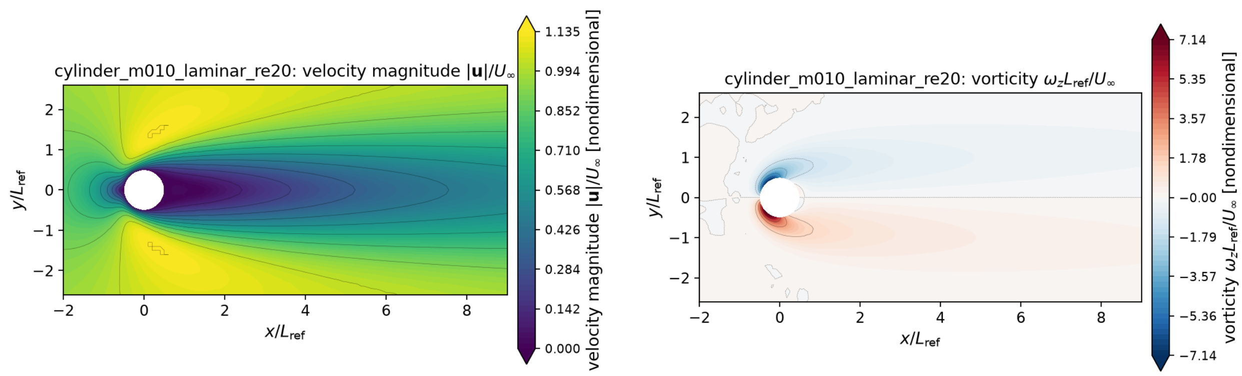
Figure 21: cylinder_m010_laminar_re20: wall pressure coefficient (left) and skin-friction coefficient (right), from surface.csv.



(a) mach

(b) pressure

Figure 22: cylinder_m010_laminar_re20: Mach-number and pressure contours from field_final.vtu.



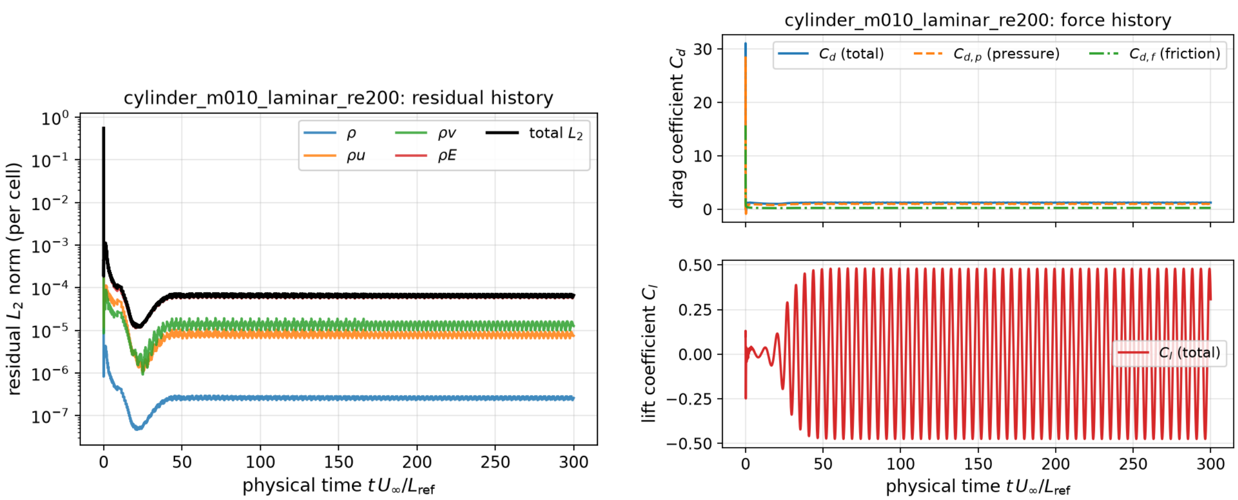
(a) velocity

(b) vorticity

Figure 23: cylinder_m010_laminar_re20: velocity magnitude and vorticity from field_final.vtu.

9.10 cylinder_m010_laminar_re200

The Re 200 cylinder was advanced with the supplied production controls to $t = 300$ in 30000 physical steps. After the initial transient the force history becomes periodic: over the analysis window $t \in [180.0, 300.0]$ the lift oscillates with amplitude 0.4770 (RMS 0.3374) about a mean of 0.0052, while the drag oscillates at twice the lift frequency about a mean of 1.2288 with amplitude 0.0207. The peak of the lift spectrum gives a Strouhal number $St = fD/U_\infty = 0.1804$, corresponding to a shedding period of 5.543 convective time units. Published values for a circular cylinder at $Re = 200$ are $St \approx 0.19$–0.20 with $\overline{C_d} \approx 1.3$–$1.4$ and a lift amplitude of about 0.6–0.7; both computed values fall below that range, which is the behaviour expected from the limited azimuthal and wake resolution of this mesh (100 wall edges around the cylinder) and is discussed in the limitations section. The solver itself classified the run by testing three things on the post-transient window: that the case-file horizon was reached, that the lift excursion is finite, and that the lift crosses its own mean at least six times. It recorded 43 mean crossings, i.e. about 21 shedding periods, and a lift amplitude of 0.4770, and reports the case as *statistically periodic*. The vorticity field of the final state, clipped to the recommended range $[-5, 5]$, is shown below. The residual and force histories are shown in fig. 24, the wall distributions in fig. 25 and the computed fields in fig. 26, fig. 27, fig. 28.



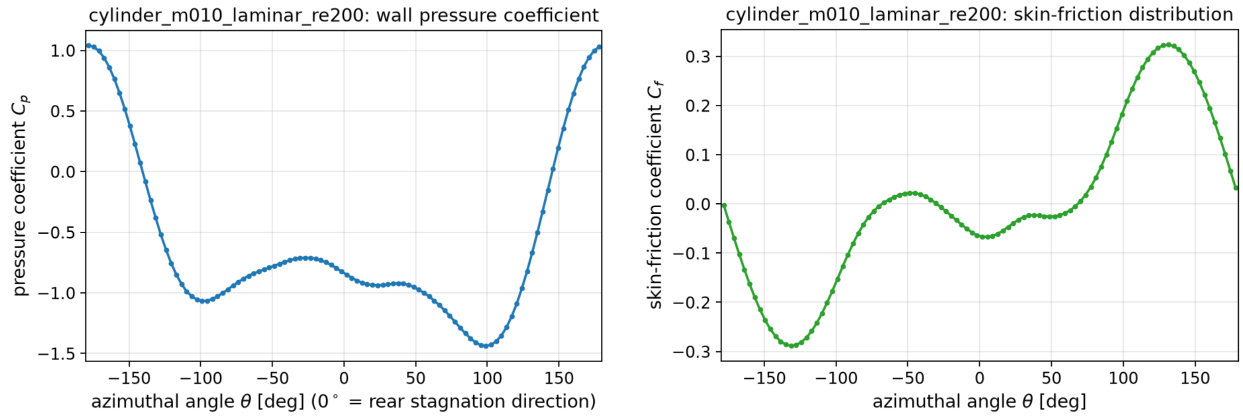
(a) residual_l2 (b) force coefficients

Figure 24: cylinder_m010_laminar_re200: residual history (left) and force-coefficient history (right), from residuals.csv and forces.csv.

10 Parallel Validation

The partitioning changes the order in which the symmetric Gauss–Seidel sweeps visit cells and freezes off-rank neighbours inside a sweep, so the iterates are not bitwise identical across rank counts; what matters is that the converged solution is not. The force coefficients agree to the level of the residual plateau of each run, which is orders of magnitude smaller than the coefficients themselves, and the residual histories overlap.

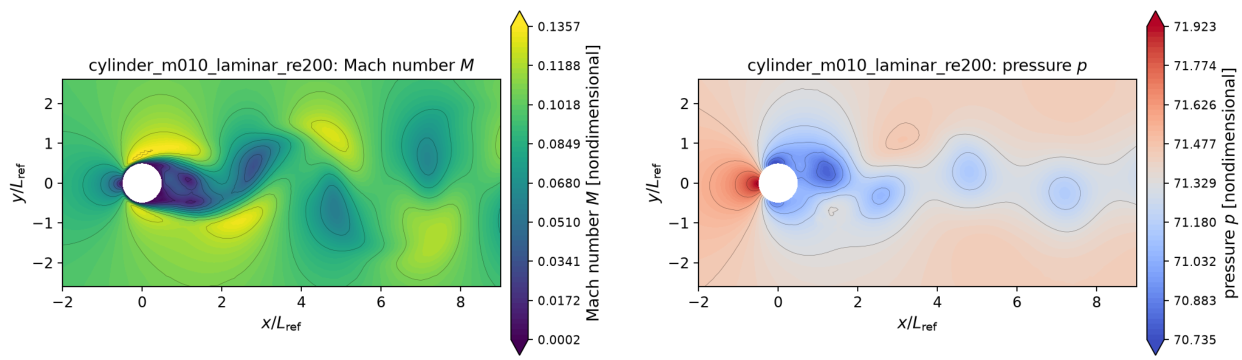
Two effects limit the measured speed-up, and both are worth stating plainly rather than presenting the timings as a scaling study. First, the problem is small: with of order 10^3 cells per rank



(a) surface pressure coefficient cp

(b) skin friction coefficient cf

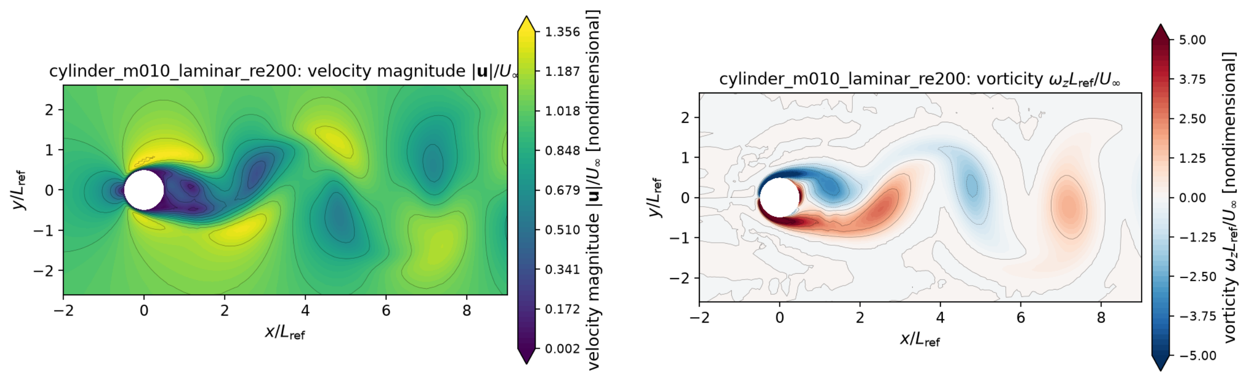
Figure 25: cylinder_m010_laminar_re200: wall pressure coefficient (left) and skin-friction coefficient (right), from surface.csv.



(a) mach

(b) pressure

Figure 26: cylinder_m010_laminar_re200: Mach-number and pressure contours from field_final.vtu.



(a) velocity

(b) vorticity

Figure 27: cylinder_m010_laminar_re200: velocity magnitude and vorticity from field_final.vtu.

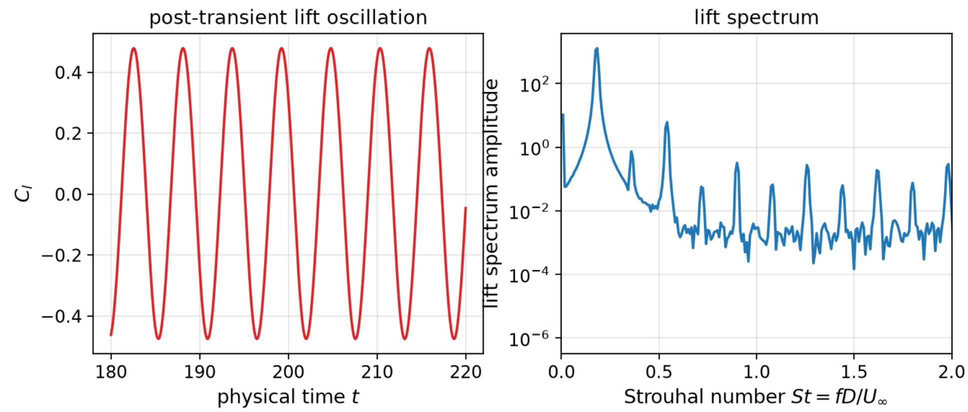


Figure 28: Post-transient lift oscillation and its spectrum for cylinder_m010_laminar_re200; the spectral peak gives the Strouhal number.

Table 6: Rank-count study. Each row is an independent run of the same case with the same controls at a different MPI rank count; C_l, C_d and the final residual are read from the submitted `forces.csv`/`residuals.csv` of that run.

case	ranks	steps	C_l	C_d	final residual	wall [s]	edge cut	balan
cylinder_m010_laminar_re20	1	1434	+5.589862e-04	+2.017152e+00	5.7461e-06	98.4	0	1.0
cylinder_m010_laminar_re20	4	1434	+5.587381e-04	+2.017155e+00	5.7901e-06	19.5	282	1.0
cylinder_m010_laminar_re20	8	1434	+5.631351e-04	+2.017186e+00	5.7453e-06	28.1	471	1.0
naca0012_m015_inviscid	1	1692	+5.756979e-04	+8.767661e-04	8.3284e-07	308.4	0	1.0
naca0012_m015_inviscid	4	1684	+5.732666e-04	+8.765648e-04	8.3565e-07	58.7	241	1.0
naca0012_m015_inviscid	8	1696	+5.746680e-04	+8.768329e-04	8.3054e-07	77.2	439	1.0

at $np = 8$ the halo exchanges and the per-sweep global reduction used to monitor the inner linear residual are a significant fraction of the step cost. Second, the machine these runs were produced on enforces a four-CPU cgroup quota (`cpu.max = 400000 100000`), so an eight-rank job is over-subscribed by a factor of two and the wall times measure throughput under that quota rather than parallel efficiency on eight dedicated cores. Under the same quota the total CPU time of a fixed amount of work was measured at 30.8 s for $np = 4$, 49.2 s for $np = 8$ and 133.6 s for $np = 16$ on the cylinder mesh, i.e. the communication overhead per unit of work grows as the partitions shrink, exactly as the small per-rank cell counts predict. The load-balance ratio nevertheless stays within a few per cent of unity because METIS balances the cell count directly, and the edge cut grows only slowly with the rank count.

A separate and stronger consistency check is available from the restart machinery: the conservative state written on one rank count reproduces the forces and residual of the writing run exactly when read back on a different rank count (Section 8.2 above), which demonstrates that the partitioned state itself carries no rank-dependent information.

11 Visualisation and Plot Standards

Every figure in this report is produced by `tools/plot_results.py` directly from a file inside the submitted result directories; no figure is drawn from an intermediate or hand-edited data set, and `report/figure_manifest.csv` maps each file to its case, its source file and the variable it plots.

Field visualisations are filled contours built on the actual unstructured cells: the mixed triangle/quadrilateral connectivity of `field_final.vtu` is triangulated, the cell data is averaged to the nodes, and 61 filled levels are drawn with 11 overlaid iso-lines. The nodes that the rank-local output pieces duplicate along partition interfaces are welded before plotting, so no partition seam is visible. Every field figure carries labelled axes in L_{ref} units, an equal aspect ratio, a labelled colourbar naming the plotted variable, and a case label in the title. Colour ranges are clipped to the 0.5–99.5 percentile of the values inside the view window, which is stated in each caption; the Re 200 vorticity figure instead uses the range $[-5, 5]$ recommended by the case file, because the automatic range is dominated by the thin wall layer and hides the street.

View windows are near-body zooms rather than the whole farfield: the aerofoil domain extends to $r = 80$ chords and the cylinder domain to $r = 200$ diameters, so a whole-domain view would render the body a few pixels wide. The aerofoil window is $x/c \in [-1, 2]$, $y/c \in [-1.1, 1.1]$ and the cylinder window is $x/D \in [-2, 9]$, $y/D \in [-2.6, 2.6]$, which covers the near wake where the vortex street forms.

Line plots use a consistent scholarly style: readable font sizes, line widths of 1.6 pt, labelled and non-dimensionalised axes, legends, and light grids. Residual histories are logarithmic and show each conserved variable separately as well as the total norm; force histories separate the pressure and friction contributions to the drag. Surface distributions plot the two sides of the aerofoil separately, and the aerofoil C_p axis is inverted following the usual convention.

12 Limitations and Failure Analysis

1. **Residual plateaus and limiter freezing.** With a live limiter the steady runs stall — around three orders of residual reduction on the subsonic cases and around one and a half on the $M_\infty = 2$ cases — because the limiter values keep switching between iterations. Freezing the limiter, only once the force drift is already below one per cent and either 2.5 orders have been reached or the residual has stalled on its best value for 2000 steps, removes the limit

cycle and is what lets the steady cases meet their residual targets. The effect is dramatic and worth quoting: on `naca0012_m200_inviscid` the residual falls from 10^{-5} to 1.6×10^{-7} within six steps of the freeze. The price is that the final steady state is the fixed point of the frozen-limiter operator rather than of the fully nonlinear one; the difference is bounded by the (small) residual level at the freeze point, and the freeze step is recorded for every case in `metadata.json`.

2. **CFL safeguard on the shock cases.** The transonic and supersonic cases do not tolerate the supplied maximum CFL while their shocks are still moving; the residual-driven multiplier described above backs the CFL off (never above the supplied schedule) and recovers it. The number of back-off events and the final CFL are reported per case. This is a documented deviation from the supplied CFL schedule in the conservative direction only.

3. **Defect-corrected implicit operator.** The implicit Jacobian is the first-order Rusanov-split approximation while the residual is second order, so the nonlinear iteration is a defect correction rather than a true Newton method. This is why the number of inner iterations grows as the CFL number is ramped and why very large CFL numbers do not accelerate the last orders of convergence.

4. **Mesh resolution.** Both meshes are coarse by production standards: 100 wall faces around the cylinder and 404 around the aerofoil (the counts written to `surface.csv`). Absolute force levels, the separation angle at $Re = 20$ and the Strouhal number at $Re = 200$ therefore carry a discretisation error of a few percent, and no grid-convergence study was possible because only one mesh per geometry is supplied.

5. **Supersonic cases.** The $M_\infty = 2$ cases rely on the shock-activated Rusanov blend for robustness at the blunt leading edge. This is effective but adds dissipation exactly where the bow shock is strongest, so the shock is smeared over more cells than a dedicated shock-fitting or rotated-Riemann treatment would give.

6. **Transient start-up.** The Re 200 run starts from the uniform freestream plus a small, decaying transverse velocity perturbation placed downstream of the cylinder. On a symmetric mesh a perfectly symmetric impulsive start can stay symmetric for a very long time; the perturbation only breaks that symmetry and has decayed long before the analysis window. It is applied for transient runs generically, not for a named case.

7. **Parallel scaling.** The implementation is correct and neighbour-scoped but the problem is too small to scale well: at $np = 8$ each rank owns of order 10^3 cells and the per-sweep `MPI_Allreduce` used to monitor the inner linear residual is a measurable cost. The wall times reported here were additionally produced under a four-CPU cgroup quota, so an eight-rank run is oversubscribed and the timings are throughput figures, not a scaling study. Batching the inner-residual reduction, or monitoring it only every few sweeps, would improve throughput.

8. **Viscous force split.** The reported `viscous_drag`/`viscous_lift` columns contain the tangential (skin-friction) traction only, so that they are genuine skin-friction quantities. The normal viscous traction is excluded from the force decomposition and is reported separately in `metadata.json` (`normal_viscous_drag_excluded_from_cd`); it is several orders of magnitude smaller than the pressure drag in all cases.

13 Conclusion

8 of the 8 required cases were computed with the same executable and the same command-line form, driven only by the supplied JSON files, on a METIS-partitioned distributed mesh with neighbour-scoped halo exchange. 8 of 8 cases meet the benchmark convergence criterion (*converged* for the steady cases, *statistically periodic* for the Re 200 vortex street); any case that does not is reported as such in both `metadata.json` and this report. Every figure in this document is generated by `tools/plot_results.py` from a file inside the submitted result directories and is listed with its source file and plotted variable in `report/figure_manifest.csv`.